

Nutritional Disorders in Adolescence and Childhood

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Objectives:

SOM.MK.III.BPM2.1.ER.2.BCHM.1076

Define and differentiate the following conditions: Nutrition, wasting, cachexia, marasmus, kwashiorkor, stunting.

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Describe malnutrition in children and its various forms.

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Describe anorexia nervosa.

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Describe childhood overnutrition including risk factors, methods of assessment, basis of management and complications.

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Define undernutrition with respect to macronutrients and micronutrients.

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Discuss the spectrum of protein-energy malnutrition including clinical features, lab findings and anthropometric measurements. Compare and contrast the following childhood nutritional disorders: Kwashiorkor, Marasmus and Marasmic-kwashiorkor.

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Explain associated micronutrient deficiency with special reference to calcium, phosphate, magnesium, iodine and selenium.

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Highlight the complications that may occur in the management of children with undernutrition and the basis for nutritional supplementation.

Notes:

Malnutrition encompasses both ends of the nutrition spectrum, from undernutrition to overnutrition. Undernutrition can result from either insufficient dietary intake or decreased ability to absorb and utilize nutrients. Nutrient deficiencies can limit growth, impair immune function, affect neurodevelopment, and increase morbidity and mortality.

Nutritional status is usually assessed using anthropometric measurements:

1. **Height-for-age** (length-for-age if < 2 years of age): Measure of skeletal and linear growth. It usually reflects the cumulative long-term impact of events affecting nutritional status. **Stunting**, refers to a child who has a low height for their age. Most often seen with **chronic** malnutrition.
2. **Weight-for-height**: a low weight-for-height usually indicates a failure to gain weight at the expected rate, which points towards **acute** malnutrition. **Wasting** is the term used for low weight-for-height.
3. **Weight-for-age**: most commonly used index of nutritional status. **Underweight** (Low

weight-for-age). A low weight-for-age has limited clinical significance since it does not distinguish between wasting and stunting.

4. **Mid-upper arm circumference (MUAC):** used to screen **wasted** children in community, outpatient or emergency settings.
5. **Body mass index (BMI) percentiles:** BMI is calculated by dividing the weight in kilograms by the square of height in meters. The problem with using BMI in childhood is that there is a change in the level of normal body fat with age. Clinicians utilize BMI percentiles that are age and gender specific for children between the ages of 2 and 20 years of age.

Severe Acute Malnutrition (SAM):

A state characterized by a weight for height ratio that is 3 standard deviations below the normal range (WHO). Worldwide about 50 million children are affected by SAM. It is common in low income countries

SAM, previously called protein energy malnutrition (PEM), manifests as a spectrum of clinical syndromes, all resulting from an inadequate dietary intake of protein and calories. At the ends of SAM spectrum are **marasmus (wasting syndrome)** and **kwashiorkor (edematous malnutrition)**. Marasmus develops when the diet is severely lacking in **calories**. Kwashiorkor occurs when **protein deprivation** is **relatively greater** than the reduction in total calories.

Pathophysiology:

Inadequate energy intake leads to various physiologic adaptations, including growth restriction; loss of fat, muscle, and visceral mass; reduced basal metabolic rate, and reduced total energy expenditure.

Metabolic changes	Early phase: rapid gluconeogenesis with resultant loss of skeletal muscle caused by use of amino acids as substrates for gluconeogenesis. Later, protein conservation phase, with fat mobilization leading to lipolysis and ketogenesis.
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	Reduced energy availability results in decreased activity of the $\text{Na}^+/\text{K}^+\text{-ATPase}$ pump resulting in sodium retention and intracellular potassium depletion .
Hormonal changes	Reduced levels of tri-iodothyroxine (T_3), which explains the reduced basal metabolic rate. Insulin is reduced, causing glucose intolerance. Glucose levels are low, with depletion of glycogen stores. Patients at risk of profound hypoglycemia during the renourishment phase. Growth hormone levels are increased, but level of its downstream effector, insulin-like growth factor 1 (IGF-1) are reduced, causing growth retardation. Leptin production is low, which may stimulate the hypothalamic-pituitary-adrenal axis to produce high levels of cortisol that contribute to lipolysis.
Immune system changes	Prone to Immune dysfunction and infections Impaired phagocytosis, and reduced secretory immunoglobulin A (IgA). Increased risk of enteric infections, which damages the mucosal lining of the GI tract and compromises absorptive capacity of intestines
Other system changes	Liver enlargement because of fatty infiltration is common in Kwashiorkor. Additionally, protein synthesis (including albumin) and gluconeogenesis (risk of hypoglycemia) are decreased. Bradycardia and hypotension are also common in the severely affected. Decreased minute ventilation, leading to impaired ventilatory response to hypoxia. Decreased neurologic global function, motor function, and memory

Clinical manifestations:

There are two functional protein compartments: the somatic compartment, (proteins in skeletal muscles), and the visceral compartment, (protein stores in the visceral organs, like the liver).

These two compartments are regulated differently. It is hypothesized that the somatic compartment is affected more severely in marasmus, and the visceral compartment is depleted more severely in kwashiorkor.

Marasmus:

There is growth retardation and **loss of muscle** mass as a result of catabolism and depletion of the **somatic protein compartment**. This seems to be an adaptive response that provides the body with amino acids as a source of energy. Marginal depletion of visceral protein compartment (critical for survival). Serum **albumin** levels are either **normal** or only slightly reduced. This can explain the **absence of edema** in marasmus.,

Due to losses of muscle and subcutaneous fat, the **extremities** are **emaciated**; by comparison, the **head appears too large for the body**. **Subcutaneous fat** is also **mobilized** and used as fuel.

Marasmus is characterized by **low weight-for-height** and **reduced MUAC**, reflecting wasting of muscle mass and depletion of body fat stores. It is the most common form of protein-energy malnutrition and is thought to be caused by inadequate intake of **all nutrients** but especially dietary energy sources (total calories).

Kwashiorkor:

Kwashiorkor occurs when protein deprivation is relatively more severe than the deficit in total calories. In Kwashiorkor, the marked protein deprivation is associated with **severe depletion** of the **visceral protein compartment** resulting in **hypoalbuminemia** giving rise to **generalized edema**. The loss of weight is masked by the increased fluid retention. To differentiate it from marasmus, there is **relative sparing** of **subcutaneous fat** and **muscle mass**.

	Marasmus	Kwashiorkor
Height and weight	Low weight for height (wasting). Low height for age (stunting), if also chronically malnourished.	Weight for age is variable. Low height for age (stunting) if also chronically malnourished
Head	Appears large relative to body, with staring eyes.	Moon face (rounded prominence of the cheeks), with or without facial edema. May appear plump.
Extremities	Emaciated arms, thighs, and buttocks, with loose folds due to loss of subcutaneous fat	Edema: Mild (only feet); Moderate (feet, legs, upper limbs); Severe (generalized and face)
Behavior	Typically irritable, fretful. Good appetite	Typically apathetic, listless; may be irritable when handled. Appetite is poor
Abdomen		Distended, dilated intestinal loops; hepatomegaly.
Skin	Thin, dry skin.	Thin, dry and peeling skin, with areas of hyperkeratosis and hyperpigmentation (dermatosis of kwashiorkor).
Hair	Thin, sparse hair that is easily plucked.	Dry, dull and hypopigmented hair, which falls out or is easily plucked. Hypopigmentation may occur in stripes if malnutrition has been intermittent ("flag sign").

Marasmic Kwashiorkor:

Mixed features of both marasmus and kwashiorkor. They have concurrent gross **wasting and edema**. Mild cutaneous and hair manifestations and an enlarged palpable fatty liver.

Anorexia Nervosa:

Anorexia nervosa is self-induced **starvation**, resulting in marked **weight loss**. It is an eating disorder that occurs primarily in previously healthy young adults who have developed an obsession with body image and thinness. Social attitudes toward body appearance, family influences, genetics, and developmental issues play a role.

DSM-5 Diagnostic criteria:

1. Restriction of energy intake that leads to a low **BMI < 18.5**
2. Intense fear of gaining weight or persistent behavior to avoid weight gain

3. Distorted perception of body weight and shape

Clinical manifestations: Similar to those in SAM.

Subtypes: Restricting subtype and the Binge-eating/purging subtype

Differential diagnosis:

1. Bulimia nervosa: binge eating and inappropriate compensatory behavior to prevent weight gain, key difference is the **BMI is not < 18.5**
2. Binge-eating disorder: recurrent binge eating with **no inappropriate compensatory behaviors**
3. Avoidant/restrictive food intake disorder: dislike of eating due to the sensory experience of eating or a fear of the consequences of eating (such as fear of vomiting or choking)

Cachexia:

Hypercatabolic state defined by a **loss of muscle mass** (with or without loss of fat) that **cannot be explained by diminished food intake**. Occurs in about 50% of **cancer** patients and is responsible for about 30% of cancer deaths.

Clinical features: Extreme weight loss and fatigue, muscle atrophy, Anemia, Edema

Mortality is a consequence of atrophy of the diaphragm and other respiratory muscles.

The inflammatory mediators, particularly TNF, IL-1, and IL-6, appear to have important roles. Cachexia also seen in chronic inflammatory conditions: Chronic infections, immunodeficiency disorders, chronic obstructive lung disease and rheumatoid arthritis

Management of Acute Severe Malnutrition:

A stepwise approach with **incremental** introduction of high calorie formula, rehydration and electrolyte correction including the hypokalemia, hypocalcemia, hypophosphatemia, and hypomagnesemia.

1. First step: Correct fluids and electrolyte abnormalities
2. Second step: Supply macronutrients by dietary therapy (milk-based formula)

The stepwise approach is important to prevent refeeding syndrome which can be fatal. What is refeeding syndrome?

- Malnourished patients have intracellular phosphate store depletion.
- On refeeding, a sudden shift from fat metabolism to carbohydrate metabolism takes place that would cause secretion of insulin to increase.
- Stimulates cellular uptake of phosphate and leads to profound hypophosphatemia
- The low phosphate will lead to low ATP, which in turn can cause rhabdomyolysis (skeletal muscle breakdown), respiratory failure, heart failure, **arrhythmias** and coma.

The same approach is considered when managing anorexia nervosa.

Overnutrition:

Obesity is by definition an accumulation of adipose tissue that is sufficient enough to impair health. The body mass index is most commonly used to measure body fat.

A BMI of more than 30 in adults is considered obese while a range between 25 and 30 is overweight. The problem is in childhood there is a change in the levels of normal body fat, it begins with high adiposity during infancy then body fat levels decrease to its lowest level followed by increase until early adulthood. Thus, BMI percentiles are used for children above the age of 2 - 20.

Obesity: A BMI above the 95th percentile

Overweight : A BMI between 85th to 95th percentile

Weight/length percentiles are used in infants younger than 2 years of age.

Overnutrition is associated with a higher risk of type 2 diabetes mellitus, cardiovascular disease, obesity, inflammatory bowel disease, and certain cancers.

Macro-mineral deficiencies:

Minerals	Function	Deficiency	Clinical Features
Calcium: Found in dairy products, legumes, broccoli, green leafy vegetables.	Building block for hydroxyapatite (for osteoid mineralization). Stabilizes resting membrane potential. Second messenger. Coagulation cascade.	Hypovitaminosis D. Hypoparathyroidism (e.g., DiGeorge). Hypomagnesemia.	Muscle cramps. Chvostek & Trousseau signs. Tetany. Seizures.
Phosphorus: Meats, eggs, dairy products, grains, legumes, and nuts; high in processed foods and sodas.	Building block for hydroxyapatite, ATP, cell membranes.	Hypovitaminosis D. Hyperparathyroidism. Refeeding syndrome.	Muscle weakness. Bone pain. Rhabdomyolysis. Osteomalacia. Respiratory failure. Seizures. Arrhythmias.

Minerals and dietary sources	Function	Deficiency	Clinical Features
Magnesium: Vegetables, cereals, nuts.	Cofactor for many enzymes including adenylate cyclase (needed for PTH activation)	Malnutrition Malabsorption Alcoholism Refeeding syndrome Diarrhea	Hypocalcemia → Neuromuscular excitability, neurologic abnormalities, ECG changes.
Potassium: Nuts, whole grains, meats, fish, beans, fruits and vegetables, especially bananas, orange juice.	Maintain resting membrane potential	Diuretics Glucosuria Hyperaldosteronism Alkalosis	Muscle weakness Rhabdomyolysis Constipation Arrhythmias Respiratory failure

Trace mineral deficiencies:

Minerals and dietary sources	Function	Deficiency	Clinical Features
Zinc: Human milk, meats, shellfish, legumes, nuts, and whole-grain cereals.	Important role in gene transcription and cell division. Important for normal spermatogenesis, fetal growth, and embryonic development.	Malabsorption. Total parenteral nutrition. Inadequate dietary intake. Bowel resection.	Alopecia. Perioral pustular skin rash. Diarrhea. Hypogonadism. Impaired wound healing. Impaired taste. Immune dysfunction.
Copper: Human milk, meats, shellfish, legumes, nuts, and whole-grain cereals.	Cytochrome (ETC). Lysyl oxidase (cross linking collagen). Iron absorption Melanin synthesis	Generalized malnutrition. Malabsorption. Prolonged diarrhea. Menke disease.	Anemia. Delayed wound healing. Brittle hair. Skin depigmentation. Muscle weakness. Osteoporosis. Ataxia, peripheral neuropathy.

Minerals	Function	Deficiency	Clinical Features
Selenium: Seafood, meats, garlic, mushrooms	Cofactor for glutathione peroxidase (protects against free radicals). Required for thyroid hormone synthesis.	Malnutrition. Total parenteral nutrition.	Skeletal muscle dysfunction → muscle pain and tenderness. Macrocytosis. Cardiomyopathy in infants and children. Thyroid dysfunction.
Iodine: Iodized salt.	Important component of thyroid hormones T₃ and T₄.	Inadequate dietary intake.	Intellectual disability, deaf mutism, spastic diplegia, and strabismus. Congenital hypothyroidism.